

COONAWARRA, Australia

WMO: 948170

Lat: 37.2906S

Lon: 140.8253E

Elev: 58

StdP: 100.64

Time Zone: 9.50 (AUA)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.6	1.8	-2.2	3.2	22.5	-0.2	3.7	13.9	11.1	12.2	10.2	12.2	1.3	110	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.1	35.6	18.4	33.2	17.9	30.8	17.3	21.0	28.1	19.7	28.0	18.7	27.2	5.2	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.1	22.3	17.4	12.5	21.4	16.0	11.5	20.5	61.2	27.5	56.7	28.2	53.3	27.4	34.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	9.0	8.1	DB	-1.7	41.1	1.1	2.2	-2.4	42.7	-3.1	43.9	-3.7	45.1	-4.5	46.7	(4)
(5)			WB	-0.6	25.4	0.7	3.5	-1.1	27.9	-1.6	29.9	-2.0	31.8	-2.5	34.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	19.5	19.5	17.3	14.2	11.9	9.7	9.3	9.9	11.4	13.3	15.5	17.4	(6)	
	DBStd	4.92	4.24	4.13	3.99	3.34	2.78	2.20	2.06	2.29	2.73	3.56	4.09	4.13	(7)	
	HDD10.0	133	0	0	0	3	12	31	39	29	13	6	1	0	(8)	
	HDD18.3	1823	36	32	70	133	202	258	280	261	210	164	108	69	(9)	
	CDD10.0	1604	293	266	227	128	69	23	17	27	53	107	165	229	(10)	
	CDD18.3	254	71	64	38	8	1	0	0	0	1	7	23	41	(11)	
	CDH23.3	3637	1010	814	528	113	6	0	0	0	8	122	370	666	(12)	
	CDH26.7	1680	519	386	236	30	0	0	0	0	0	39	151	320	(13)	
(14) Wind	WSAvg	4.1	4.3	4.1	3.9	3.4	3.6	3.6	4.2	4.3	4.4	4.4	4.2	4.2	(14)	
(15) Precipitation	PrecAvg	643	31	22	30	42	58	85	91	88	72	49	38	36	(15)	
	PrecMax	876	105	56	81	91	136	174	143	163	131	98	94	116	(16)	
	PrecMin	423	2	0	4	4	7	16	43	26	30	2	13	6	(17)	
	PrecStd	107	28	15	19	22	31	36	28	38	27	28	19	28	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.8	37.2	36.1	30.3	23.6	18.0	16.7	20.6	24.5	31.0	33.9	37.6	(19)	
		MCWB	18.9	19.2	18.1	16.2	13.3	12.7	11.8	12.7	15.1	15.7	17.9	18.1	(20)	
	2%	DB	35.8	34.4	32.4	26.5	20.3	15.9	15.0	17.8	21.0	26.7	30.9	33.8	(21)	
		MCWB	18.8	18.4	17.3	14.9	13.1	11.8	11.1	11.8	13.6	15.1	16.4	17.8	(22)	
	5%	DB	32.5	31.6	29.2	23.6	18.3	14.8	14.0	15.9	18.8	23.4	27.8	30.2	(23)	
		MCWB	18.4	18.1	16.6	14.3	13.0	11.4	10.7	11.2	12.6	14.2	16.2	17.1	(24)	
	10%	DB	29.1	28.5	25.8	20.9	16.6	13.8	13.0	14.3	16.9	20.5	24.4	26.9	(25)	
		MCWB	17.4	17.6	16.2	13.8	12.6	10.9	10.3	10.5	12.1	13.5	15.6	16.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.9	22.4	20.8	17.7	15.8	14.7	13.3	14.1	16.3	19.8	21.1	22.1	(27)	
		MCDB	28.2	31.0	27.7	23.4	18.6	15.6	14.9	18.6	21.6	22.0	28.6	27.3	(28)	
	2%	WB	20.7	20.6	18.9	16.5	14.7	12.8	12.2	12.6	14.5	16.5	19.0	19.4	(29)	
		MCDB	28.9	28.4	27.3	21.9	17.5	14.7	14.0	15.9	19.2	22.2	26.7	27.1	(30)	
	5%	WB	19.5	19.3	17.9	15.6	13.9	12.0	11.4	11.7	13.4	15.1	17.6	18.1	(31)	
		MCDB	29.2	27.8	25.6	20.8	16.6	13.9	13.1	14.7	17.6	21.6	24.8	26.4	(32)	
	10%	WB	18.2	18.3	17.0	14.7	13.2	11.3	10.7	11.0	12.4	14.0	16.2	16.9	(33)	
		MCDB	27.4	26.7	24.3	19.4	15.9	13.2	12.5	13.7	16.1	19.6	23.1	25.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.1	14.3	13.7	12.5	9.3	8.3	7.6	8.6	9.9	11.8	13.1	14.6	(35)	
		MCDBR	20.9	19.3	19.0	17.2	11.5	9.4	8.0	10.6	12.9	16.9	19.0	21.1	(36)	
		MCWBR	6.5	6.2	6.3	7.2	6.2	6.2	5.2	6.1	6.9	7.6	7.2	7.4	(37)	
	5% WB	MCDBR	16.5	15.3	15.1	12.8	8.6	7.7	7.1	9.2	11.1	13.9	15.2	16.4	(38)	
		MCWBR	7.2	6.1	6.2	6.3	5.3	5.7	5.3	5.7	6.6	7.5	6.8	7.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.338	0.333	0.320	0.312	0.306	0.302	0.301	0.316	0.334	0.354	0.336	0.335	(40)	
	taud		2.499	2.524	2.558	2.565	2.564	2.562	2.557	2.500	2.441	2.381	2.471	2.495	(41)	
	Ebn at Noon		996	977	951	898	843	816	839	878	919	944	991	1003	(42)	
	Edh at Noon		113	106	94	82	71	67	71	86	104	121	116	115	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.46	6.49	5.09	3.53	2.35	1.92	2.07	2.94	4.08	5.44	6.51	7.28	(44)	
	RadStd		0.41	0.32	0.22	0.22	0.12	0.14	0.11	0.15	0.33	0.42	0.41	0.44	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	N/A	-96	+132	+39

Nomenclature: See separate page